

# Overpriced?

An Analysis of the Modern NFL Quarterback Salary

James Barnhill October 31, 2024

# Introduction

## James Barnhill

- Physics Background
  - B.S. Davidson College
  - M.S. University of California, Davis
- Current Role
  - High School Physics Teacher at Dixon High School

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- NFL & Pittsburgh Steelers Fan

# Research Problem

## Are quarterbacks overvalued?

- Context:
  - Star quarterbacks can make over 20% of an NFL team's salary cap<sup>1</sup>
  - The job of an offensive line is to protect the quarterback from injury
  - Losing a quarterback to a season-ending injury can be extremely detrimental to an NFL team's season

1. A historical table of quarterback contracts can be found at <https://overthecap.com/contract-history/quarterback>.

# Research Problem

## Are quarterbacks overvalued?

- Questions:
  - Are large quarterback contracts signed at the expense of an effective offensive line?
  - Do lower paid offensive lines increase the risk of quarterback injury?
  - **Can an analysis of NFL player salaries be used to predict team performance (win-loss records)?**

# Research Problem

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  - Do lower paid offensive lines increase the risk of quarterback injury?
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- Hypothesis: ***There is a statistically significant correlation between NFL position salaries and win-loss records.***

# Analysis

## Data Collected

- Data for each NFL team from 2014-2023
  - Win-Loss Record
  - Strength of Schedule
  - Number of weeks that the usual starting quarterback missed due to injury
  - Total salaries grouped by each position on the roster, represented as a percentage of the team's total spending for that year

# Analysis

## 3 Models

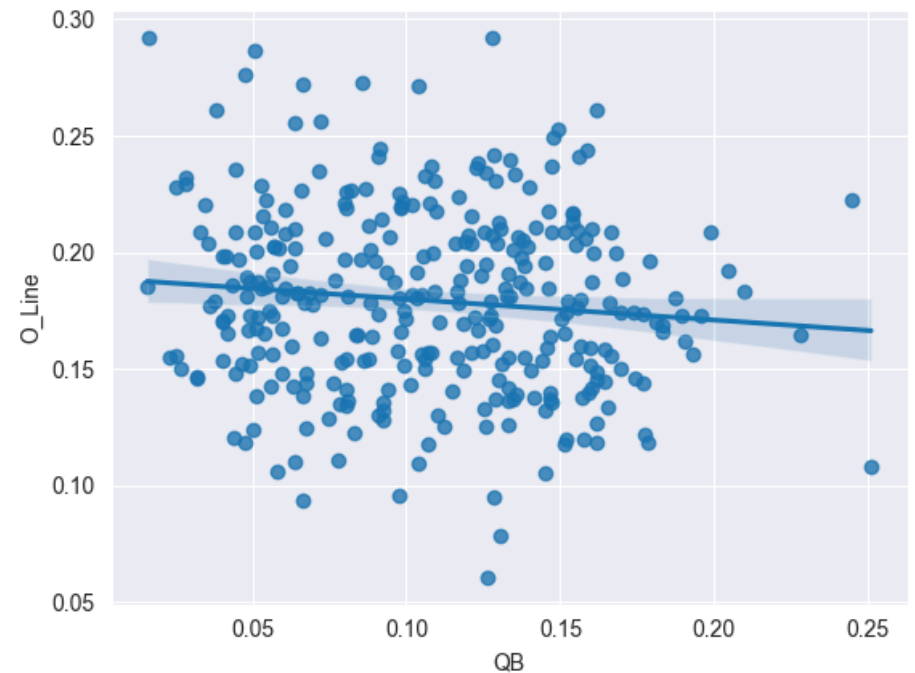
- Are quarterback salaries correlated with offensive line salaries?
  - Linear Model
- Is losing a quarterback to significant injury time correlated with offensive line salaries?
  - Logistic Model
- Can a model be created to predict NFL season records from position salary data?
  - Linear Model



# Results

## Model 1

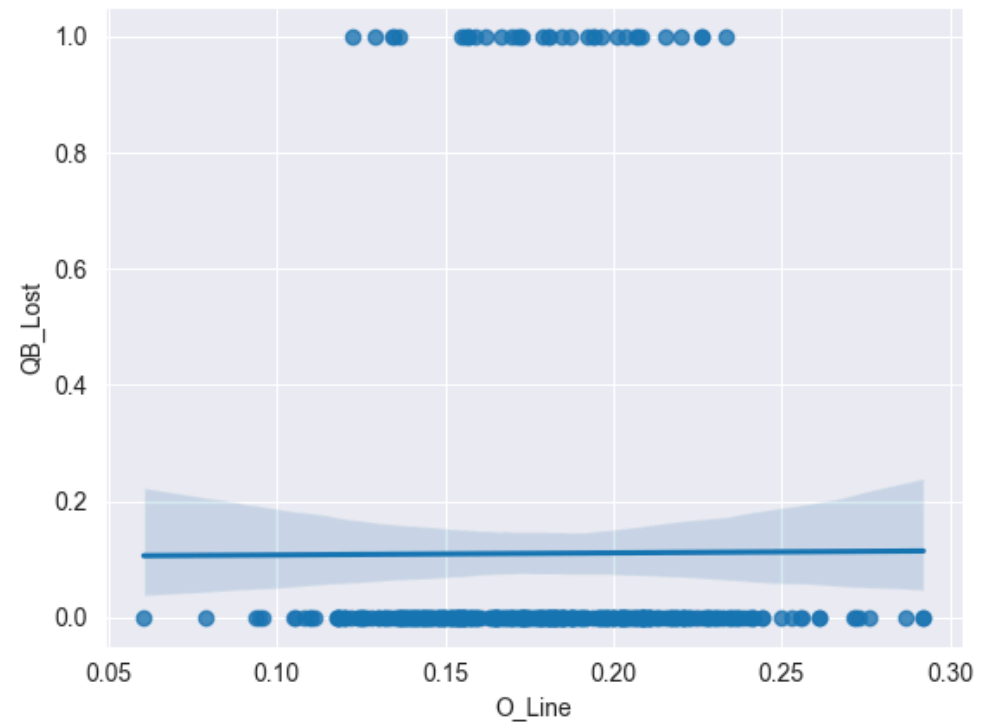
- **Are quarterback salaries correlated with offensive line salaries?**
- No significant correlation detected



# Results

## Model 2

- **Is losing a quarterback to significant injury time correlated with offensive line salaries?**
- No significant correlation detected
- Model unable to predict a single injury on the test set



# Results

## Model 3

- Can a model be created to predict NFL season records from position salary data?
- No significant correlation detected
- Model used to predict season records for 2024

|    | Team                  | Predicted_Record |
|----|-----------------------|------------------|
| 0  | Seattle Seahawks      | 0.675            |
| 1  | Jacksonville Jaguars  | 0.671            |
| 2  | Chicago Bears         | 0.639            |
| 3  | Miami Dolphins        | 0.637            |
| 4  | New England Patriots  | 0.617            |
| 5  | Las Vegas Raiders     | 0.603            |
| 6  | Washington Commanders | 0.600            |
| 7  | Buffalo Bills         | 0.589            |
| 8  | Cincinnati Bengals    | 0.588            |
| 9  | Atlanta Falcons       | 0.576            |
| 10 | Philadelphia Eagles   | 0.554            |

# Limitations

- Not enough data points for a very robust model
- More in depth data needed
  - Play time
  - Injury status

# Proposed Actions

## More Data!

- Consider salaries individually
  - Play time
  - Injury status
- Consider correlations between salaries and on-field statistics
  - Offensive line vs sacks allowed
  - Quarterback vs passing yards

# Benefits

## Goal: Create a successful model

- Strategically budget salary cap by position
- Optimize money spent on each position
- Find undervalued players to be able to afford high valued players
- 'Moneyball' in the NFL<sup>2</sup>

2. An explanation of moneyball in baseball can be found at <https://eutawstreetreport.com/the-impact-of-moneyball-in-mlb/>.